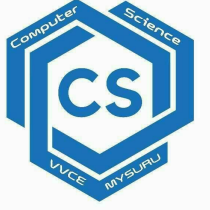


Vidyavardhaka College of Engineering, Mysuru

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Blockchain Technology-BCSBT603 Module 2: Decentralization

7rd SEM

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Our Vision: "VVCE shall be a leading Institution in engineering and management education enabling individuals for significant contribution to the society"

Topics to be covered....

- Decentralization
- ❖ Decentralization using Blockchain.
- ❖ Blockchain and full ecosystem decentralization.
- ❖ Decentralization
- ❖ Smart contract
- ❖ Decentralization organizations.
- ❖ Decentralization autonomous organizations
- ❖ Decentralization autonomous Corporation
- ❖ Decentralization autonomous societies
- ❖ Decentralization platform

▪ **Cryptography and Technical Foundation:**

- ❖ Mathematics
- ❖ Cryptography
- ❖ Cryptographic primitives
- ❖ Symmetric Cryptography
- ❖ DES
- ❖ AES

Decentralization

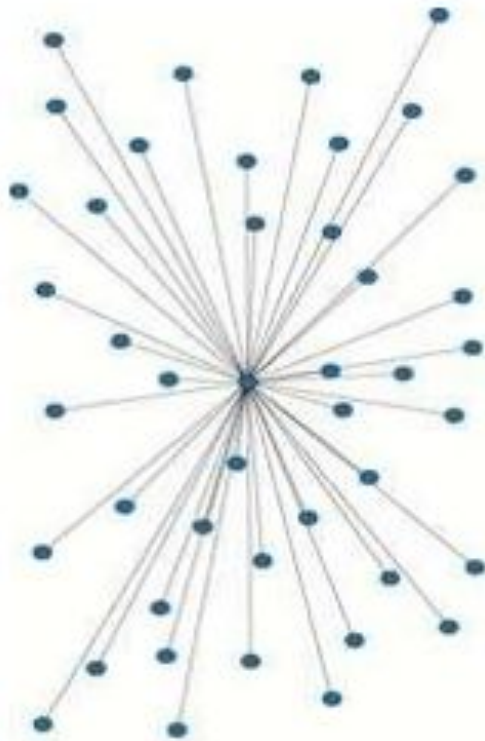
- The basic idea of decentralization is to distribute control and authority to peripheries instead of one central authority being in full control of the organization.
- This results in several benefits for organizations, such as
 - Increased efficiency,
 - Quicker decision making,
 - A reduced burden on top management.

Decentralization using Blockchain

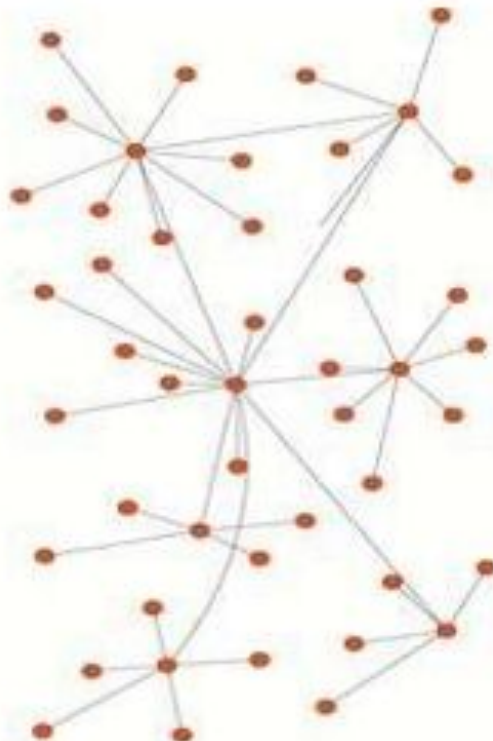
- . Decentralization is a core benefit and service provided by the blockchain technology.
- . Blockchain by design is a perfect vehicle for providing a platform that does not need any intermediaries and can function with many different leaders chosen via consensus mechanisms.
- . This model allows anyone to compete to become the decision-making authority.
- . This competition is governed by a consensus mechanism and the most commonly used method is known as Proof of Work (PoW)

- Decentralization is applied in varying degrees from semi-decentralized to fully decentralized depending on the requirements and circumstances.
- Decentralization can be viewed from a blockchain perspective as a mechanism that provides a way to remodel existing applications and paradigms or build new applications in order to give full control to users.

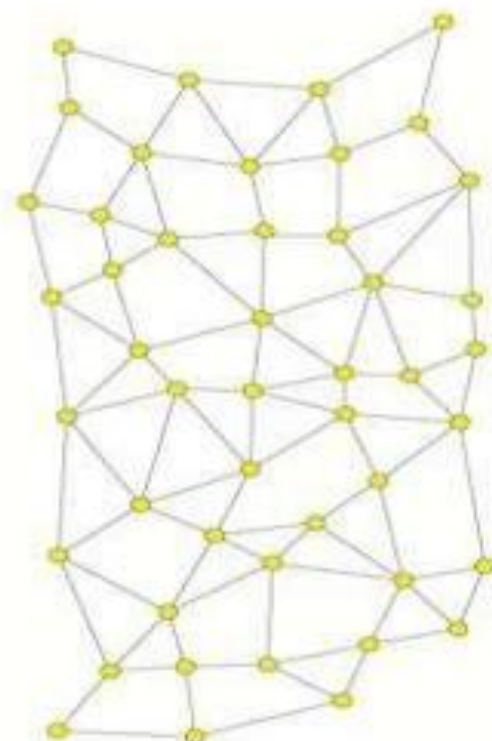
Different types of networks/systems



Centralized network



Decentralized network



Distributed network

1. **Centralized systems** are conventional (client-server) IT systems in which there is a single authority that controls the system, and who is solely in charge of all operations on the system.

- All users of a centralized system are dependent on a single source of service.
- The majority of online service providers, including Google, Amazon, eBay, and Apple's App Store, use this conventional model to deliver services.
- **Facebook,**

Facebook operates on a centralized model where all user data, posts, and interactions are stored on Facebook's servers.

2. Distributed systems are a computing paradigm whereby two or more nodes work with each other in a coordinated fashion in order to achieve a common outcome and it's modeled in such a way that end users see it as a single logical platform.

- Even if some of the nodes become faulty or network links break, the distributed system should tolerate this and should continue to work flawlessly in order to achieve the desired result.
- **General Examples** :Telephone and Cellular system, Wireless sensor networks, Routing algorithm, Peer to peer networks.
- **Blockchain-Based Social Media (e.g., Steemit)**
- Steemit is a blockchain-based social media platform where data is stored across multiple nodes in a distributed ledger.
- No single entity owns or controls the content, and it remains available as long as the blockchain network exists.

3. Decentralized means no node is instructing any other node as to what to do.

- The code runs on a peer-to-peer network of nodes and no single node has control over the dApp.
- Depending on the functionality of the dApp, different data structures can be used to store the application data.
- Bitcoin uses a blockchain decentralized ledger of transactions.
- This enables a user to agree on something via a consensus algorithm without the need for a central trusted third party, intermediary, or service provider
- **Decentralized System: Mastodon**
 - Mastodon is a decentralized social media platform where different servers (called instances) operate independently.
 - Users can create accounts on different instances, but they can still interact with users on other instances.

Methods of Decentralization

- Two methods that can be used to achieve decentralization

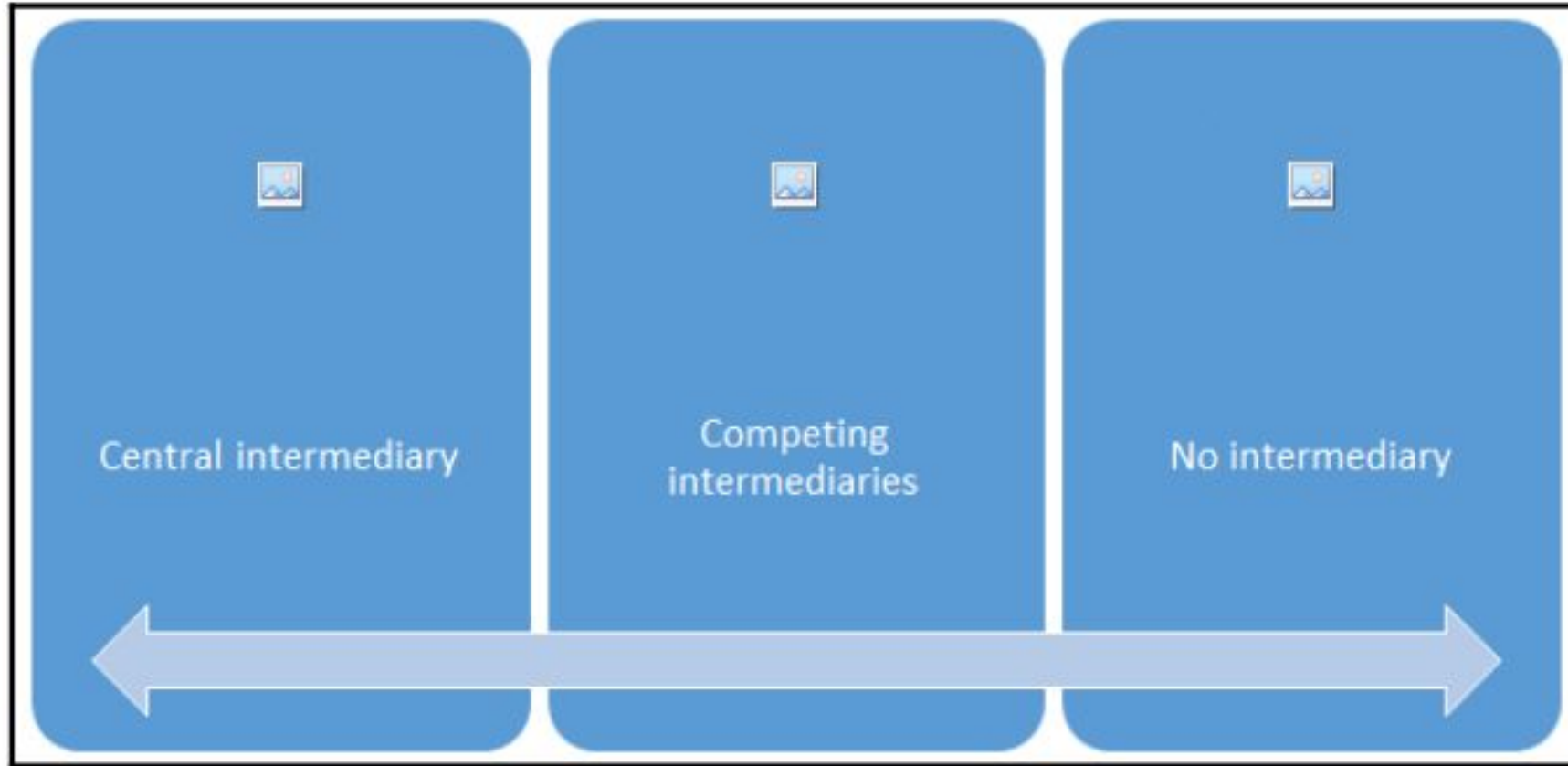
- **1. Disintermediation**

- With blockchain technology, it is possible to send this money directly to your friend without the need for a bank.
- All you need is the address of your friend on the blockchain.
- This way, the intermediary is no longer required and decentralization is achieved by disintermediation.

- **2. Through competition**

- A group of service providers compete with each other in order to be selected for the provision of services by the system.
- Smart Contracts can choose external data providers from a large numbers of providers based on reputation or voting, previous score, review and quality of service.

Scale of decentralization



1. Central Intermediary (Fully Centralized)

- A single entity acts as the middleman, controlling all transactions, communications, or processes.
- Users must trust this central authority for access and decision-making.
- **Example:**
 - Banks controlling financial transactions.
 - Facebook controlling social media interactions.
 - Centralized cloud providers like Google Drive or Dropbox.

2. Competing Intermediaries (Partially Decentralized)

- Multiple intermediaries exist, allowing users to choose between different providers.
- Each intermediary still has control over its own network, but users have alternatives.
- Some level of competition and redundancy reduces risks but doesn't eliminate intermediaries.
- **Example:**
 - Multiple banks offering financial services (Visa, Mastercard, PayPal).
 - Different cloud service providers (AWS, Google Cloud, Microsoft Azure).
 - Federated systems like Mastodon, where different servers (instances) operate independently.

3. No Intermediary (Fully Decentralized or Distributed)

- No middleman—users interact directly with each other.
- Trust is established through cryptographic mechanisms or consensus algorithms.
- **Example:**
 - Bitcoin and blockchain networks (users transact directly via peer-to-peer mechanisms).
 - IPFS (InterPlanetary File System) for decentralized file storage.
 - DAOs (Decentralized Autonomous Organizations), where governance is handled by smart contracts.

Routes to decentralization

- Ethereum, which is currently the tool of choice of many developers for building decentralized application.

How to decentralize

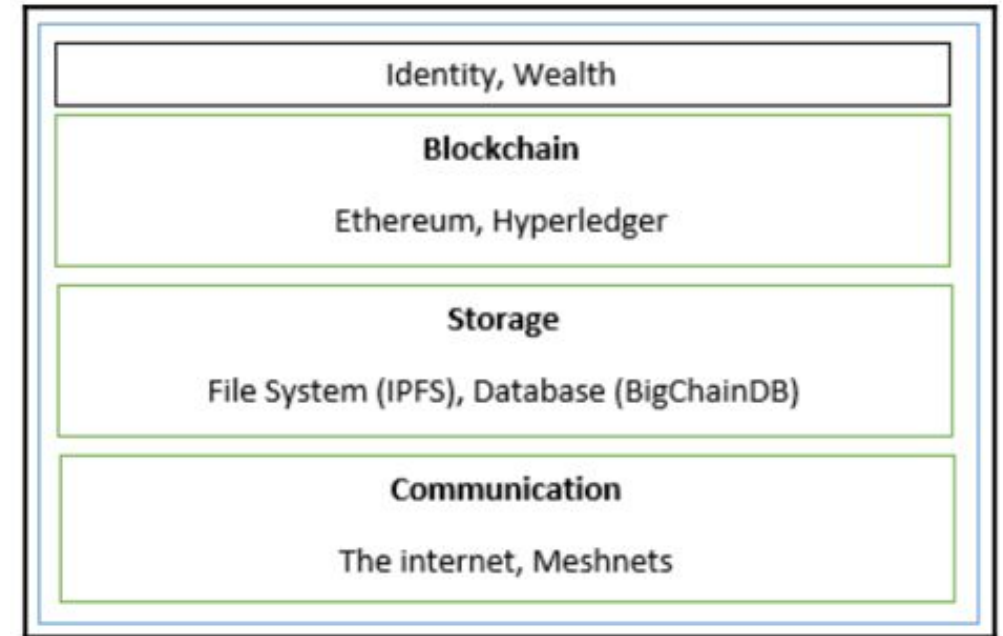
1. What is begin decentralized ?
2. What level of decentralization is required ?
3. What blockchain is used ?
4. what security mechanism is used ?

Use case : Money transfer

1. Answer 1 : Money transfer system
2. Answer 2: Disintermediation.
3. Answer 3: Bitcoin.
4. Answer 4: Atomicity.

Blockchain and full ecosystem decentralization

- In order to achieve complete decentralization, it is necessary that the environment around the blockchain is also decentralized.
- Blockchain itself is a distributed ledger that runs on top of conventional systems. These elements include storage, communication, and computation.



- **1. Storage**

- Data can be stored directly in a blockchain,

- It does achieve decentralization,

- **Major disadvantage:** not suitable for storing large amounts of data by design.

- A better alternative is to use distributed hash tables (DHTs). DHTs were originally used in peer-to-peer file sharing software, such as BitTorrent, Napster, Kazaa, and Gnutella.
 - **Two main requirements:** high availability and link stability, which means that data should be available when required and network links should also always be accessible.
 - Inter Planetary File System (IPFS) possesses both of these properties
 - The vision is to provide a decentralized World Wide Web by replacing the HTTP protocol.
 - IPFS uses Kademlia DHT and merkle DAG (Directed Acyclic Graph) to provide the storage and searching functionality.

2. **Communication**

- It is generally considered that the Internet (the communication layer in blockchain) is decentralized.
- This is true to some extent as the original vision of the Internet was to develop a decentralized system.
- ISP.

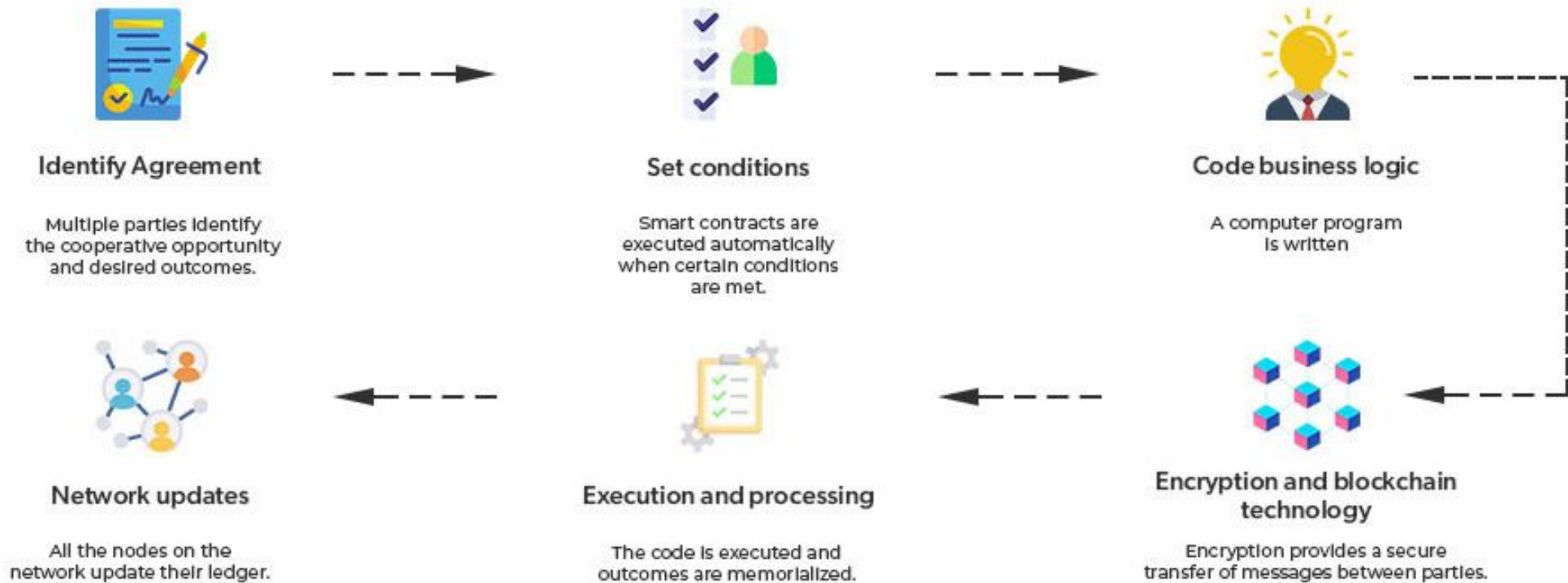
3. **Computation :**

- Decentralization of computing or processing is achieved by a blockchain technology such as Ethereum, where smart contracts with embedded business logic can run on the network.
- Other blockchain technologies also provide similar processing layer platforms where business logic can run over the network in a decentralized manner.
- storage layer uses technologies such as Interplanetary File Systems (IPFS) and BigChainDB to enable decentralization

Smart contract

- A smart contract can be thought of as a small decentralized program.
- Smart contracts do not necessarily need a blockchain to run; however, due to the security benefits that the blockchain technology provides, it is now becoming almost a standard to use blockchain as a decentralized execution platform for smart contracts.
- A smart contract usually contains some business logic and a limited amount of data.
- Actors or participants in the blockchain use these smart contracts or they run autonomously on behalf of the network participants
- Small programs reside on the blockchain and execute business logic if some specific criteria are met.

How does a Smart Contract Work?



Elements/ Components

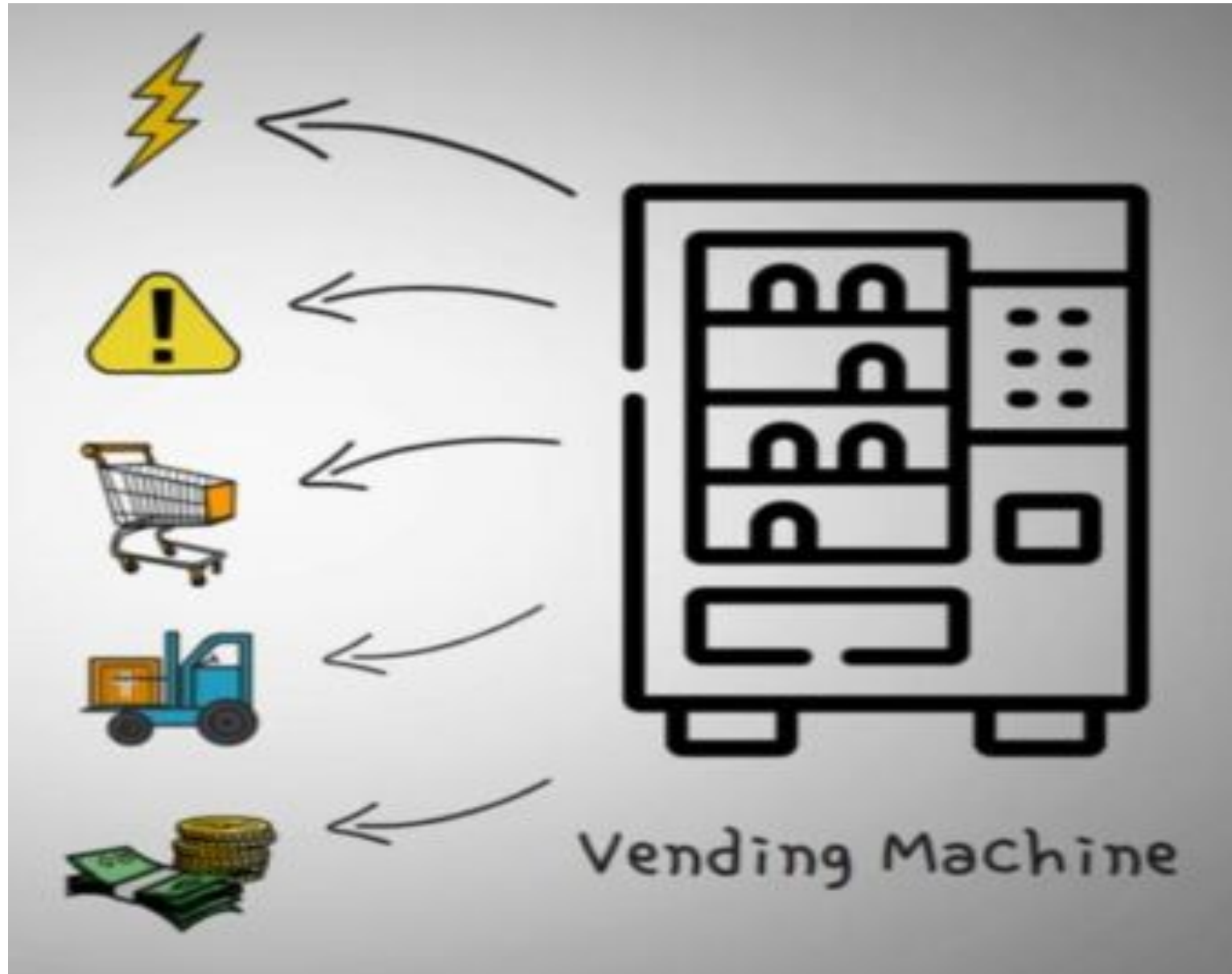
- Decentralized organizations (DOs). – S/W
- Decentralized Autonomous organizations (DAOs) - Fully Automated
- Decentralized Autonomous Corporations(DACs) – subset of DAO
- Decentralized Autonomous Societies (DASs) – Societies

Decentralized organizations

- **Decentralized organization (DOs)** are software programs that run on a blockchain and are based on the idea of real human organizations with people and protocols.
- Once a DO, in the form of a smart contract or a set of smart contracts, is added to the blockchain, it becomes decentralized and parties interact with each other based on the code defined within the DO software.

Example: GitHub Open-Source Projects

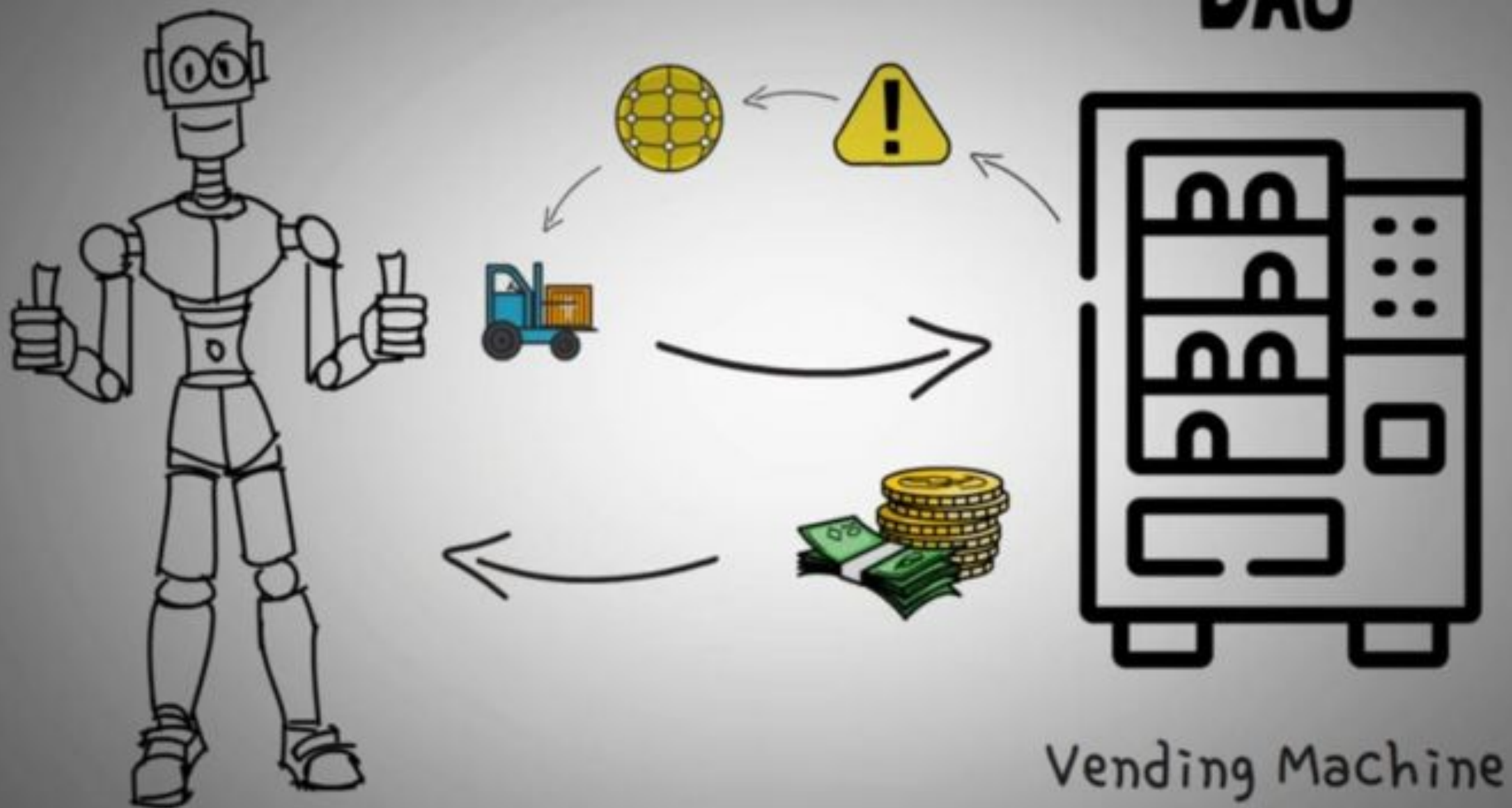
- Many open-source projects on GitHub function as decentralized organizations. Developers from around the world collaborate, contribute code, and make decisions collectively, often without a formal hierarchy. Decisions are typically made through discussions, proposals, and voting among contributors.



Decentralized autonomous organizations

- **Decentralized autonomous organization (DAO)** is also a computer program than runs on top of a blockchain and embedded within it are governance and business logic rules.
- DAO and DO are basically the same thing, but the main difference is that DAOs are autonomous, which means that they are fully automated and contain artificially intelligent logic, whereas DOs lack this feature and rely on human input in order to execute business logic.

DAO



Decentralized autonomous corporations

- **DAOs, Decentralized autonomous corporations (DACs)** are a similar concept but are considered a smaller subset of DAOs.
- The definitions of DACs and DAOs can sometimes overlap, but a general difference is that DAOs are usually considered to be nonprofit, whereas DACs can make money via shares offered to the participants and by paying dividends.
- These corporations can run a business automatically without human intervention based on the logic programmed within them.

Decentralized autonomous societies

- **Decentralized Autonomous Society (DAS)** is an advanced concept where entire communities or societies operate on decentralized, blockchain-based governance principles.
- Many services that a government offers can be delivered via blockchain, such as Government Identity Card systems, passport issuance, and records of deeds, marriages, and births.

Decentralized applications

- All DAOs, DACs, and DOs are basically decentralized applications that run on top of a blockchain in a peer-to-peer network.
- Decentralized applications or DAPPs are software programs that can run on their own blockchain, use another already existing established blockchain, or use only protocols of an existing blockchain solution.
- These are called Type I, Type II, and Type III DAPPs.

• Requirements of a Decentralized applications (DAPPs)

1. The DAPP should be fully open source and autonomous and no single entity should be in control of a majority of its tokens. All changes to the application must be consensus-driven based on the feedback given by the community.
2. Data and records of operations of the application must be cryptographically secured and stored on a public, decentralized blockchain in order to avoid any central points of failure.
3. A cryptographic token must be used by the application in order to provide access and rewards to those who contribute value to the applications, for example, miners in bitcoin.
4. The tokens must be generated by the decentralized application according to a standard cryptographic algorithm. This generation of tokens acts as a proof of the value to contributors (for example, miners).

Operations of a DAPP :

- Proof of Work and Proof of Stake can be used to establishment the DAPP
- DAPP can distribute tokens (coins) via **mining, fundraising, and development.**
- DAPP - Examples
 - **KYC-Chain** - manage Know Your Customer (KYC) data in a **secure and convenient** way based on smart contracts.
 - **OpenBazaar** - commercial activities b/w **buyer and seller** using **peer-peer**, uses **DHTs**, It makes use of bitcoin as a payment network
 - **Lazooz** – smart transportation; It allows peer-to-peer ride sharing and users can be incentivized by proof of movement

KYCChain

- **KYCChain** is a blockchain-based platform designed to streamline **Know Your Customer (KYC)** processes for businesses, financial institutions, and governments.
- It helps companies verify user identities in a secure, decentralized, and efficient manner while ensuring compliance with global regulatory standards.

How KYCChain Works:

- **User Submits Identity Data** → Uploads documents like passport, driver's license, or biometrics.
- **Data is Verified & Stored on Blockchain** → Ensures tamper-proof security.
- **User Grants Access to Businesses** → Users control who can access their KYC data.
- **Business Verifies KYC Data** → Instantly validates customer identity without redoing the KYC process.

OpenBazaar: A Decentralized Marketplace

- **OpenBazaar** was a **decentralized, peer-to-peer (P2P) marketplace** that allowed users to buy and sell goods and services directly, without intermediaries like Amazon or eBay.
- It leveraged **Bitcoin and other cryptocurrencies** for transactions, offering censorship-resistant and fee-free e-commerce.

How OpenBazaar Worked:

- **Download & Install OpenBazaar** – Users run the software on their own devices.
- **Set Up a Store or Browse Listings** – Sellers listed products, and buyers searched for items.
- **Make a Purchase with Cryptocurrency** – Transactions occurred directly between buyer and seller.
- **Use Escrow for Security (Optional)** – A trusted third party could resolve disputes.
- **Complete the Trade** – Once confirmed, funds were released to the seller.

Lazooz: A Decentralized Ride-Sharing Platform

Lazooz was a blockchain-based, **decentralized ride-sharing platform** that aimed to compete with traditional services like Uber and Lyft but without a central authority.

It used a **cryptocurrency-based reward system** to incentivize drivers and passengers, promoting a **collaborative, peer-to-peer (P2P) economy**.

How Lazooz Worked

- 1. Users Installed the Lazooz App** – The app tracked movement and rewarded users with **Zooz tokens**.
- 2. Drivers & Riders Connected P2P** – No central authority managed ride requests.
- 3. Payments in Zooz Tokens** – Transactions were made securely on the blockchain.
- 4. Community Governance** – Users had a say in the platform's future developments.

Platforms for decentralization

- many platforms available for decentralization
- Many companies introduced platforms that promise to make distributed application development easy, accessible, and secure for users.
- **Ethereum**
 - Ethereum tops the list as being the first blockchain that introduced a Turing-complete language and the concept of a virtual machine.
 - scripting language in bitcoin and many other cryptocurrencies.
 - Currency tokens on Ethereum are called Ethers.

Maidsafe

- Maidsafe provides a SAFE (Secure Access for Everyone) network, made up of unused computing resources, such as storage, processing power, and the data connections of its users.

How the SAFE Network Works

- **Users Provide Resources** – People contribute unused storage, bandwidth, and processing power.
- **Data is Encrypted & Split** – Files are broken into chunks, encrypted, and distributed across the network.
- **Users Access Data Privately** – Only the rightful owner (with cryptographic keys) can retrieve the files.
- **SafeCoin Incentives** – Users who contribute resources earn SafeCoin (now SN Token).

Lisk (LSK): A Blockchain for JavaScript-Based dApps:

Lisk (LSK) is a **blockchain platform** designed to make it easier for developers to build and deploy **decentralized applications (dApps)** using **JavaScript and TypeScript**.

Unlike Ethereum, which requires Solidity, Lisk provides a **Sidechain Development Kit (SDK)** to create independent blockchains connected to the Lisk mainchain.

How Lisk Works

- 1. Developers Use Lisk SDK** – Create dApps using JavaScript/TypeScript.
- 2. Deploy on a Sidechain** – Each app runs on its independent blockchain, ensuring flexibility.
- 3. Secure the Network with DPoS** – Users stake LSK tokens to vote for trusted delegates.
- 4. Seamless Interaction with Lisk Mainchain** – dApps can interact with the main Lisk blockchain if needed.

Cryptography and Technical Foundations

- Cryptography is the science of making information secure in the presence of adversaries.
- Cryptography provides various security services, such as **Confidentiality, Integrity, Authentication**, (Entity Authentication and Data origin authentication), **accountability** and **nonrepudiation**.

Mathematics

- Cryptography relies on a variety of mathematical concepts to secure data and communications.
- **sets** and **groups** are fundamental concepts in **abstract algebra**, and they play a crucial role in modern cryptography, particularly in public-key cryptosystems.

1. Sets

- A **set** is a collection of distinct elements. Sets are foundational in many areas of mathematics, including cryptography.
- **Examples of Sets in Cryptography**
- **Set of Natural Numbers** $N=\{1,2,3,\dots\}$
- **Set of Integers** $Z=\{\dots,-2,-1,0,1,2,\dots\}$
- **Set of Prime Numbers** $P=\{2,3,5,7,11,\dots\}$ $P = \{2, 3, 5, 7, 11, \dots\}$
- **Set of Elements in Modular Arithmetic** (e.g., $Z_n=\{0,1,\dots,n-1$

Application in Cryptography:

- The **set of prime numbers** is used in RSA encryption.
- The **set of elements in modular arithmetic** is used in Diffie-Hellman key exchange

•2. Groups

- A **group** is a mathematical structure consisting of a set with an operation that satisfies certain properties.

Definition of a Group

A group $(G, *)$ consists of:

1. **Closure:** If $a, b \in G$, then $a * b \in G$.
2. **Associativity:** $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.
3. **Identity Element:** There exists an element $e \in G$ such that $a * e = e * a = a$ for all $a \in G$.
4. **Inverse Element:** For every $a \in G$, there exists an element a^{-1} such that $a * a^{-1} = a^{-1} * a = e$.

Application in Cryptography

- **RSA Encryption:** Uses the multiplicative group $(\mathbb{Z}_n^*, \cdot)$
- **Diffie-Hellman Key Exchange:** Uses cyclic groups in modular arithmetic.
- **Elliptic Curve Cryptography (ECC):** Uses the group of points on an elliptic curve.

Field in Cryptography

- A **field** is an algebraic structure that extends the concept of a group by incorporating two operations: **addition** and **multiplication**.
- Fields play a crucial role in cryptography, particularly in **finite field arithmetic**, which is used in **AES encryption**, **Elliptic Curve Cryptography (ECC)**, and **error correction codes**.

1. Definition of a Field

A **field** $(F, +, \cdot)$ is a set of elements with two operations (addition and multiplication) that satisfy the following properties:

Addition Properties (Forms an Abelian Group)

1. **Closure:** If $a, b \in F$, then $a + b \in F$.
2. **Associativity:** $(a + b) + c = a + (b + c)$.
3. **Commutativity:** $a + b = b + a$.
4. **Identity Element:** There exists an element 0 such that $a + 0 = a$.
5. **Inverse Element:** For each $a \in F$, there exists $-a$ such that $a + (-a) = 0$.

Multiplication Properties (Forms an Abelian Group Except for Zero)

6. **Closure:** If $a, b \in F$, then $a \cdot b \in F$.
7. **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
8. **Commutativity:** $a \cdot b = b \cdot a$.
9. **Identity Element:** There exists an element $1 \neq 0$ such that $a \cdot 1 = a$.
10. **Inverse Element:** For each $a \neq 0$, there exists a^{-1} such that $a \cdot a^{-1} = 1$.

Distributive Property

11. $a \cdot (b + c) = a \cdot b + a \cdot c$.

Example

Question: Is $(\mathbb{Z}, +)$ a group?

Solution:

$\mathbb{Z} = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$ is an abelian group.

CAIN Property	Explanation
Closure	If $a, b \in G$, then $(a \bullet b) \in G$. If $a = 5, b = -2 \in \mathbb{Z}$ then $(a + b) = -3 \in \mathbb{Z}$
Associative	$a \bullet (b \bullet c) = (a \bullet b) \bullet c$ for all $a, b, c \in G$. $5 + (3 + 7) = (5 + 3) + 7 \in \mathbb{Z}$
Identity element	$(a \bullet e) = (e \bullet a) = a$ for all $a \in G$. $(5 + 0) = (0 + 5) = 5$ for all $a \in G$.
Inverse element	$(a \bullet a') = (a' \bullet a) = e$ for all $a, a' \in G$. $(5 + -5) = (-5 + 5) = 0$ for all $5, -5 \in \mathbb{Z}$
Commutative	$(a \bullet b) = (b \bullet a)$ for all $a, b \in G$. $(5 + 9) = (9 + 5)$ for all $9, 5 \in \mathbb{Z}$.

□ Order

- This is the number of elements in a field. It is also known as the cardinality of the field.

□ Prime fields

- This is a finite field with a prime number of elements.
- It has specific rules for addition and multiplication, and each nonzero element in the field has an inverse.
- Addition and multiplication operations are performed modulo p .

Rings

- A **ring** is an algebraic structure that generalizes the concept of numbers and provides a foundation for many cryptographic algorithms. Rings are particularly important in **lattice-based cryptography**, **homomorphic encryption**, and **code-based cryptography**.

A ring R denoted by $\{R, +, *\}$, is a set of elements with two binary operations, called addition and multiplication, such that for all $a, b, c \in R$ the following axioms are obeyed:

- ❖ Group (A1-A4), Abelian Group(A5).
- ❖ Closure under multiplication (M1): If $a, b \in R$ then $ab \in R$
- ❖ Associativity of multiplication (M2): $a(bc) = (ab)c$ for all $a, b, c \in R$
- ❖ Distributive laws (M3) :

$$a(b + c) = ab + ac \text{ for all } a, b, c \in R$$

$$(a + b)c = ac + bc \text{ for all } a, b, c \in R$$

□ A cyclic group:

- A cyclic group is a type of group that can be generated by a single element called the group generator.

Notationally: If G is a cyclic group and g is a generator, every element $x \in G$ can be written as:

- $x = g^n$ (if the group operation is multiplication)
- $x = ng$ (if the group operation is addition) where n is an integer.



Examples:

1. **Finite Example:** The set $\{1, 2, 3, 4\}$ under addition modulo 4 is a cyclic group. Here, the element 1 is the generator, because:
 - $1 + 1 \equiv 2 \pmod{4}$
 - $1 + 1 + 1 \equiv 3 \pmod{4}$
 - $1 + 1 + 1 + 1 \equiv 0 \pmod{4}$, and the set repeats.

□ An abelian group:

- An abelian group is formed when the operation on the elements of a set is commutative.
- Commutative law basically means that changing the order of the elements does not affect the result of the operation, for example, $A \times B = B \times A$.

□ Modular arithmetic:

- Modular arithmetic is a fundamental mathematical concept used in cryptography, particularly in RSA encryption, Diffie-Hellman key exchange, Elliptic Curve Cryptography (ECC), and hash functions.

1. Definition of Modular Arithmetic

Modular arithmetic involves calculations with numbers **wrapped around** after reaching a certain value, called the **modulus** n .

For a number a , its **modulo** n is:

$$a \bmod n = \text{remainder when } a \text{ is divided by } n.$$

Formally, we say two numbers a and b are **congruent modulo** n if:

$$a \equiv b \pmod{n}$$

which means $a - b$ is divisible by n .

Cryptography

□ Confidentiality

- Confidentiality is the assurance that information is only available to authorized entities.

□ Integrity

- Integrity is the assurance that information is modifiable only by authorized entities.

□ Authentication

- Authentication provides assurance about the identity of an entity or the validity of a message.
There are two types of authentications,

1. Entity authentication

2. Data origin authentication

Entity authentication

- Entity authentication is the assurance that an entity is currently involved and active in a communication session.
- Traditionally, users are issued a username and password, which are used to gain access to the platforms they are using. This is called single factor authentication as there is only one factor, namely *something you know*, that is, the password and username.
- This type of authentication is not very secure due to various reasons, such as password leakage;
- The use of additional techniques for user identification is known as **multifactor authentication** or **two-factor authentication**.
- Various factors are described here:
 1. The first factor is something you have, such as a hardware token or smart card.
 2. The second factor is something you are, which uses biometric features in order to identify the user.

Data origin authentication

- The source of information is verified.
- Data origin authentication implies data integrity because if a source is corroborated, then data must not have been altered. Various methods, such as **Message Authentication Codes (MACs)** and **Digital signatures** are most commonly used.

Digital Signatures in Cryptography

- A **digital signature** is a cryptographic technique used to verify the authenticity, integrity, and non-repudiation of a message or document. It ensures that a message has not been tampered with and that it comes from the rightful sender.

How Digital Signatures Work

- A digital signature involves three main steps:

1. Key Generation:

1. The sender generates a **public key (PK)** and a **private key (SK)**.
2. The **private key** is used for signing.
3. The **public key** is used for verification.

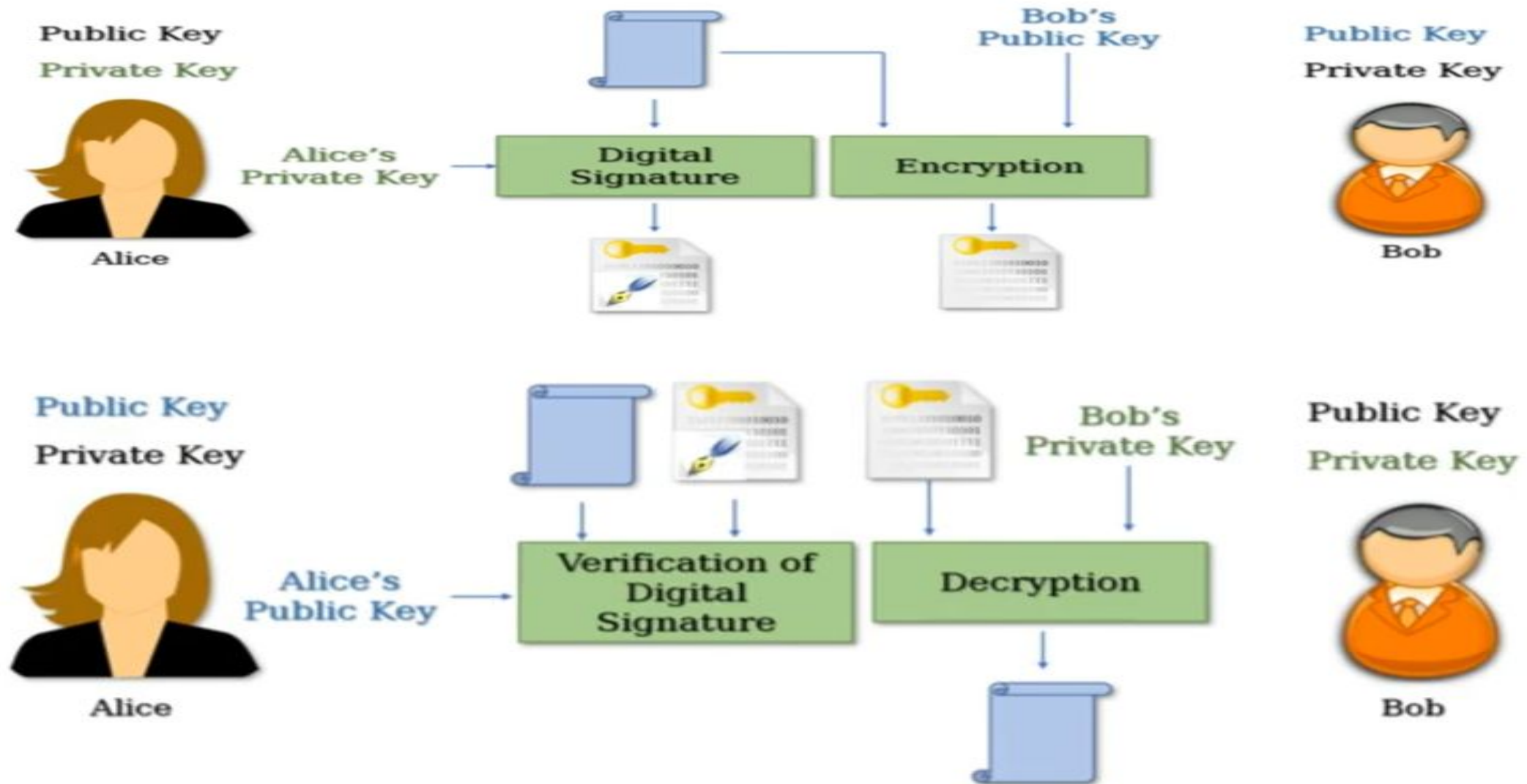
2. Signing Process:

1. The sender hashes the message to create a **message digest**.
2. The hash is encrypted using the sender's **private key** to create the **digital signature**.

3. Verification Process:

1. The receiver decrypts the signature using the **sender's public key** to obtain the original hash.
2. The receiver hashes the received message and compares it with the decrypted hash.
3. If both hashes match, the signature is **valid**.

Digital Signature



Message Authentication Codes (MACs) in Cryptography

- A **Message Authentication Code (MAC)** is a cryptographic technique used to ensure the **integrity** and **authenticity** of a message.
- Unlike digital signatures, which use public-key cryptography, MACs use **symmetric-key cryptography**, meaning the sender and receiver share a **secret key**.

How it Works

- A MAC takes three inputs:
- **Message (M)** – The data that needs authentication.
- **Secret Key (K)** – Shared between the sender and receiver.
- **MAC Function (MAC_K)** – Computes a fixed-length authentication tag.
- The sender computes:

$$\text{MAC} = \text{MAC_K}(\text{M})$$

sends **(M, MAC)** to the receiver.

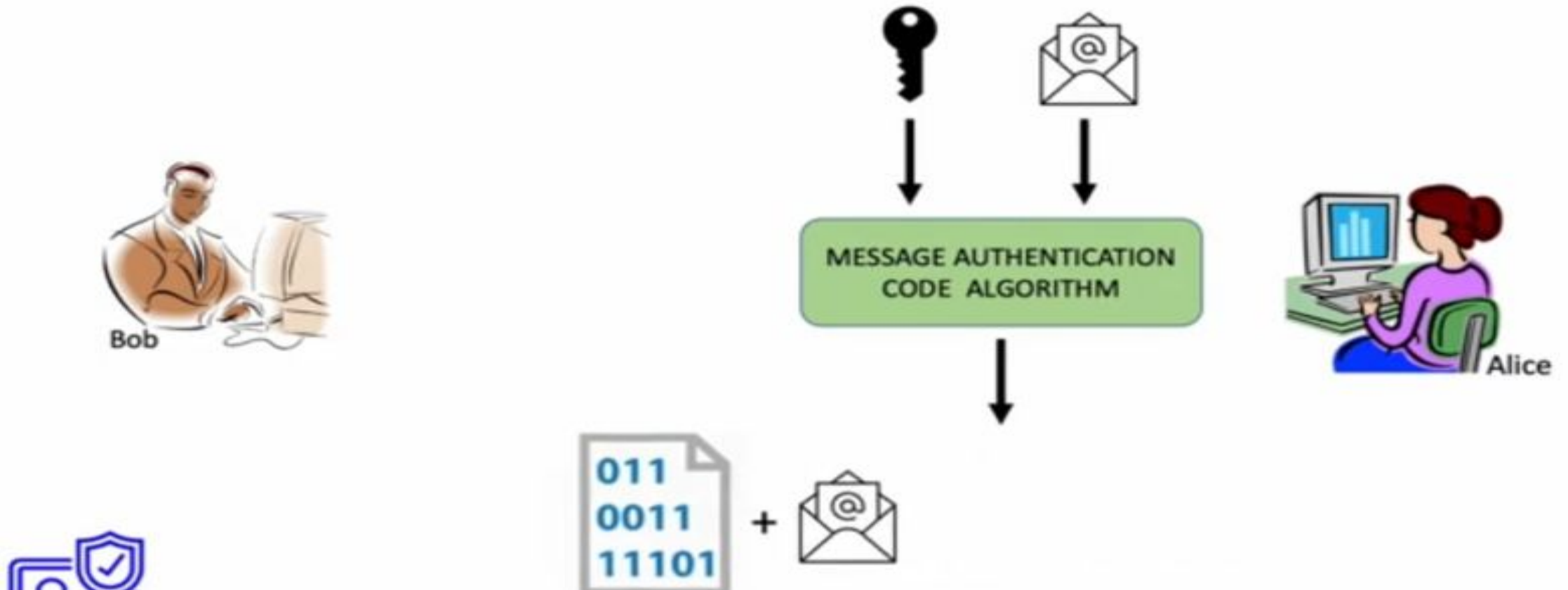
- The receiver also computes:

$$\text{MAC}' = \text{MAC_K}(\text{M})$$

- If **MAC' = MAC**, the message is **authentic** and **unchanged**.

Message Authentication Codes

MESSAGE AUTHENTICATION CODE(MAC)



Non-repudiation

- The non-repudiation protocol usually runs in a communication network and is used to provide evidence that an action has been taken by an entity (originator or recipient) on the network.
- There are two communication models that can be used to transfer messages from originator A to recipient B:
 1. Message is sent directly from originator A to recipient B.
 2. Message is sent to a delivery agent from originator A, which then delivers the message to recipient B.
- The main requirements of a non-repudiation protocol are fairness, effectiveness, and timeliness.

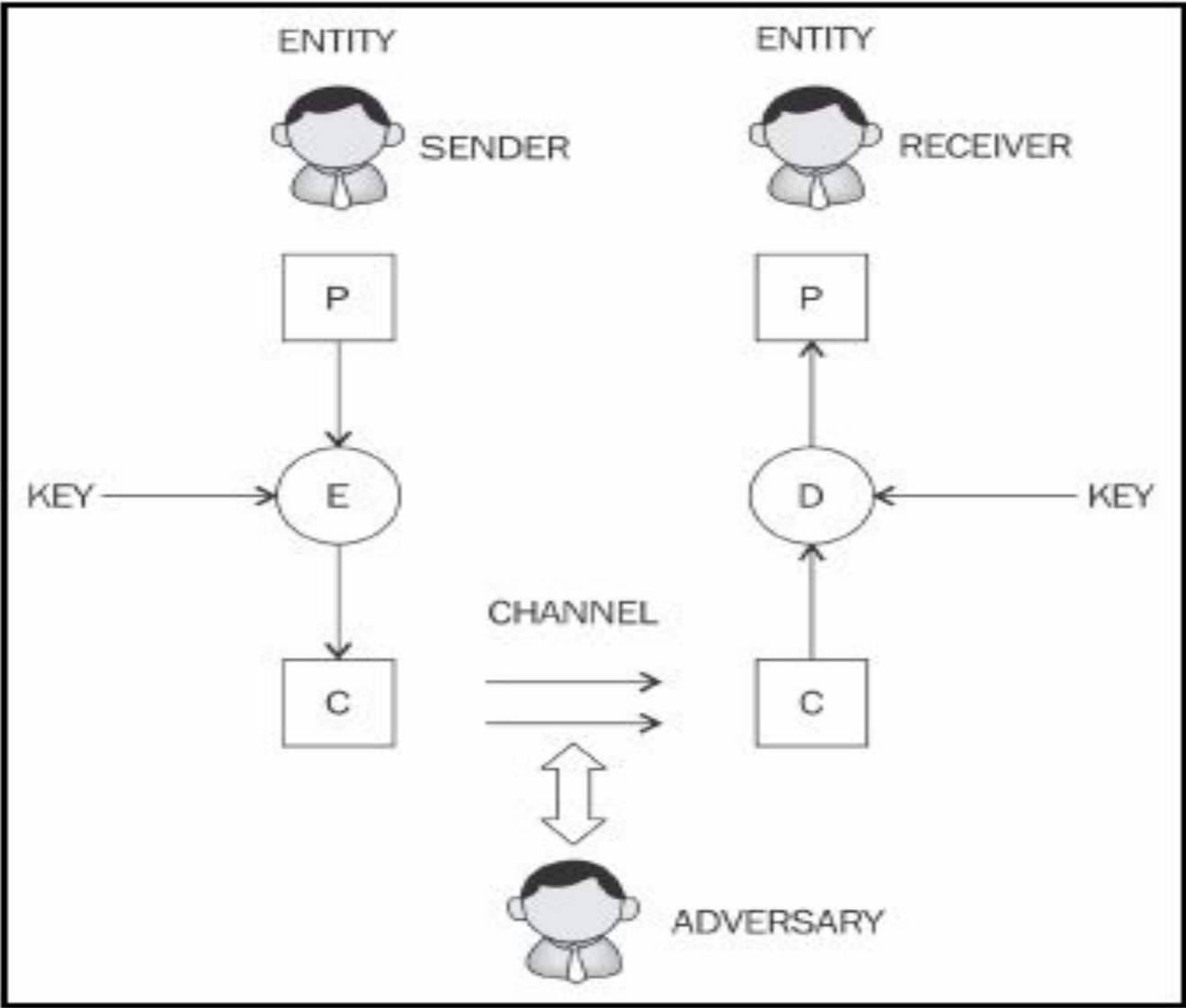
Accountability

- Accountability is the assurance that actions affecting security can be traced to the responsible party.
- This is usually provided by logging and audit mechanisms in systems where a detailed audit is required due to the nature of the business, for example, in electronic trading systems.
- Detailed logs are vital to trace an entity's actions,
- For example, when a trade is placed in an audit record with the date and time stamp and the entity's identity is generated and saved in the log file.
- This log file can optionally be encrypted and can be part of the database or a standalone ASCII text log file on a system.

Cryptographic primitives

- Cryptographic primitives are the basic building blocks of a security protocol or system.
- cryptographic algorithms that are essential for the building of secure protocols and systems.
- A **security protocol** is a set of steps taken in order to achieve required security goals by utilizing appropriate security mechanisms.
- Various types of security protocols are in use, such as authentication protocols, nonrepudiation protocols, and key management protocols.

A generic cryptography model is shown in the following diagram:



A model showing the generic encryption and decryption model

- **Entity:** It is either a person or a system that sends, receives, or performs operations on data
- **Sender:** Sender is an entity that transmits the data
- **Receiver:** Receiver is an entity that takes delivery of the data
- **Adversary:** This is an entity that tries to circumvent the security service
- **Key:** A key is some data that is used to encrypt or decrypt data
- **Channel:** Channel provides a medium of communication between entities

Cryptography is mainly divided into two categories, namely symmetric and asymmetric cryptography.

□ **Symmetric cryptography :**

- Symmetric cryptography refers to a type of cryptography whereby the key that is used to encrypt the data is the same for decrypting the data, and thus it is also known as a shared key cryptography.
- The key must be established or agreed on before the data exchange between the communicating parties. This is the reason it is also called **secret key cryptography**.
- There are two types of symmetric ciphers, stream ciphers and block ciphers.
- **Data Encryption Standard (DES)** and **Advanced Encryption Standard (AES)** are common examples of block ciphers, whereas RC4 and A5 are commonly used stream ciphers.

Stream ciphers

- These ciphers are encryption algorithms that apply encryption algorithms on a bit-by-bit basis to plain text using a key stream.
- There are two types of stream ciphers: synchronous and asynchronous.
- **Synchronous stream ciphers** are ones where key stream is dependent only on the key.

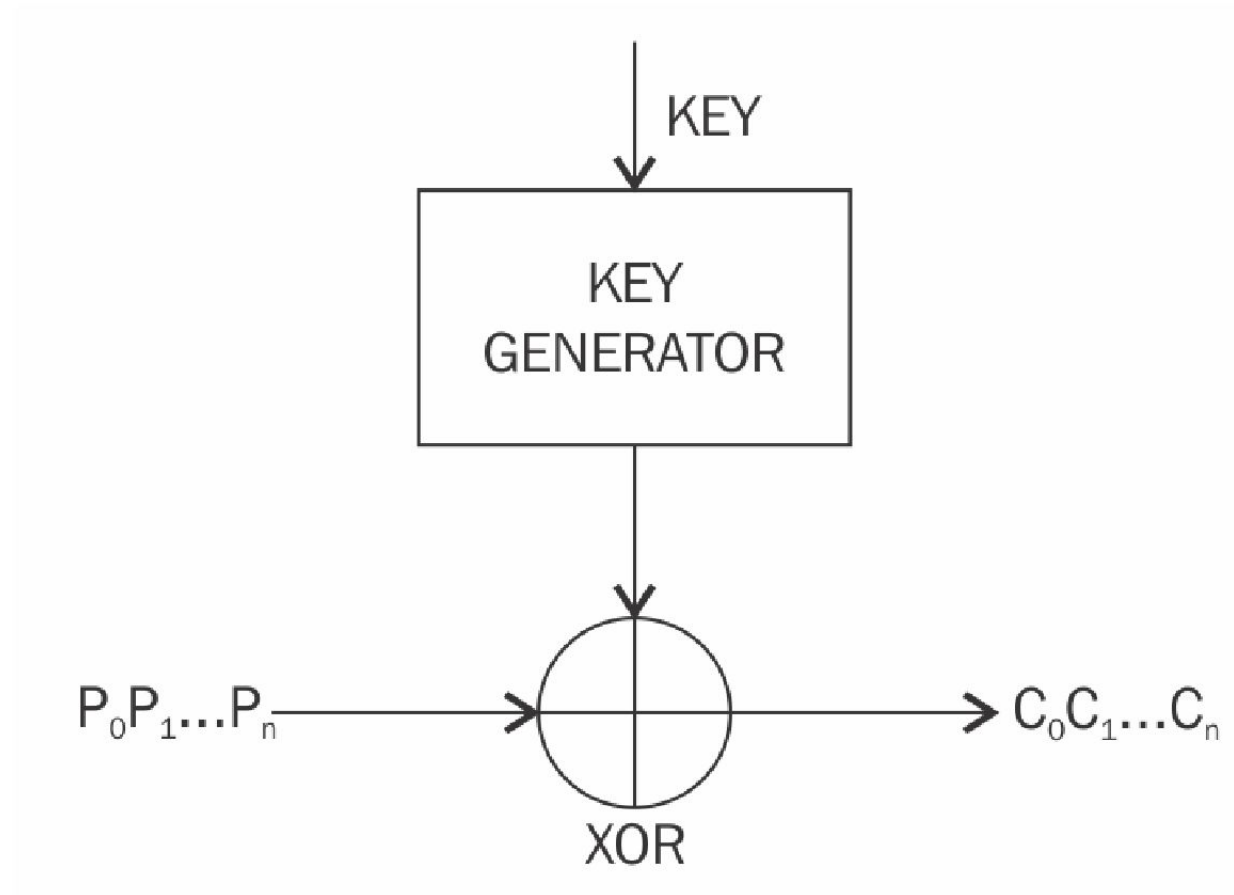
Example: **RC4 (Rivest Cipher 4)**

- **Asynchronous stream ciphers** have a key stream that is also dependent on the encrypted data.

Example: **CFB (Cipher Feedback Mode) using AES**

- In stream ciphers, encryption and decryption are basically the same function because they are simple modulo 2 additions or XOR operation.
- The key requirement in stream ciphers is the security and randomness of key streams.

Operation of stream cipher



$P = P_1P_2P_3, \dots$
 $C_1 = E_{k1}(P_1)$

$C = C_1C_2C_3, \dots$
 $C_2 = E_{k2}(P_2)$

$K = (k_1, k_2, k_3, \dots)$
 $C_3 = E_{k3}(P_3) \dots$

Example for Stream Cipher

- Let's encrypt "**HELLO**" using a simple stream cipher concept (assuming a simplified keystream).
- **Plaintext (ASCII):**

H (72) E (69) L (76) L (76) O (79)

Keystream (Example values):

12 34 56 78 90

Ciphertext (XOR operation):

$72 \oplus 12 = 68$ $69 \oplus 34 = 103$ $76 \oplus 56 = 116$ $76 \oplus 78 = 2$ $79 \oplus 90 = 21$

Ciphertext Output (Decimal):

68 103 116 2 21

Block ciphers

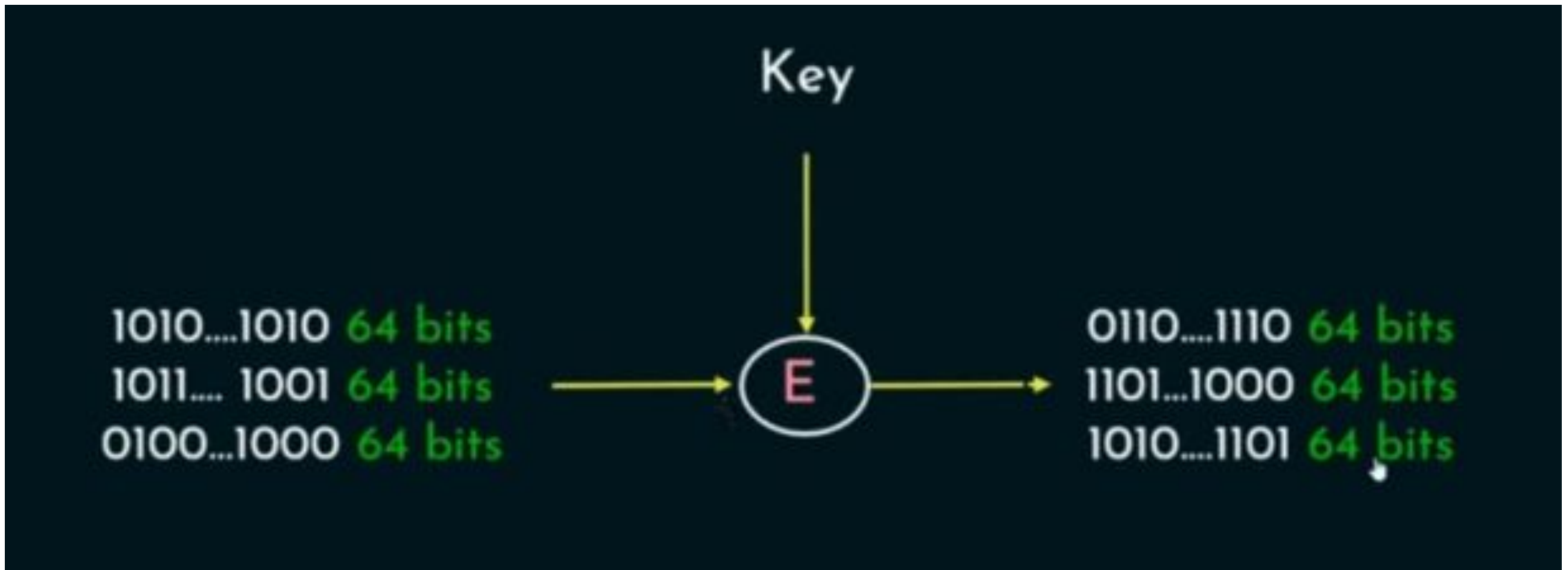
- These are encryption algorithms that break up a text to be encrypted (plain text) into blocks of fixed length and apply encryption block by block.
- Common block sizes are 64 bits (DES) and 128 bits (AES).
- Block ciphers are usually built using a design strategy known as Feistel cipher.
- Recent block ciphers, such as AES (Rijndael) have been built using a combination of substitution and permutation called **substitution permutation network (SPN)**.
- The Feistel Ciphers structure is based on the idea of combining multiple rounds of repeated operations to achieve desirable cryptographic properties known as **confusion and diffusion**.

□ **Confusion:** DES uses S-boxes (Substitution Boxes) in the round function to replace bits unpredictably.

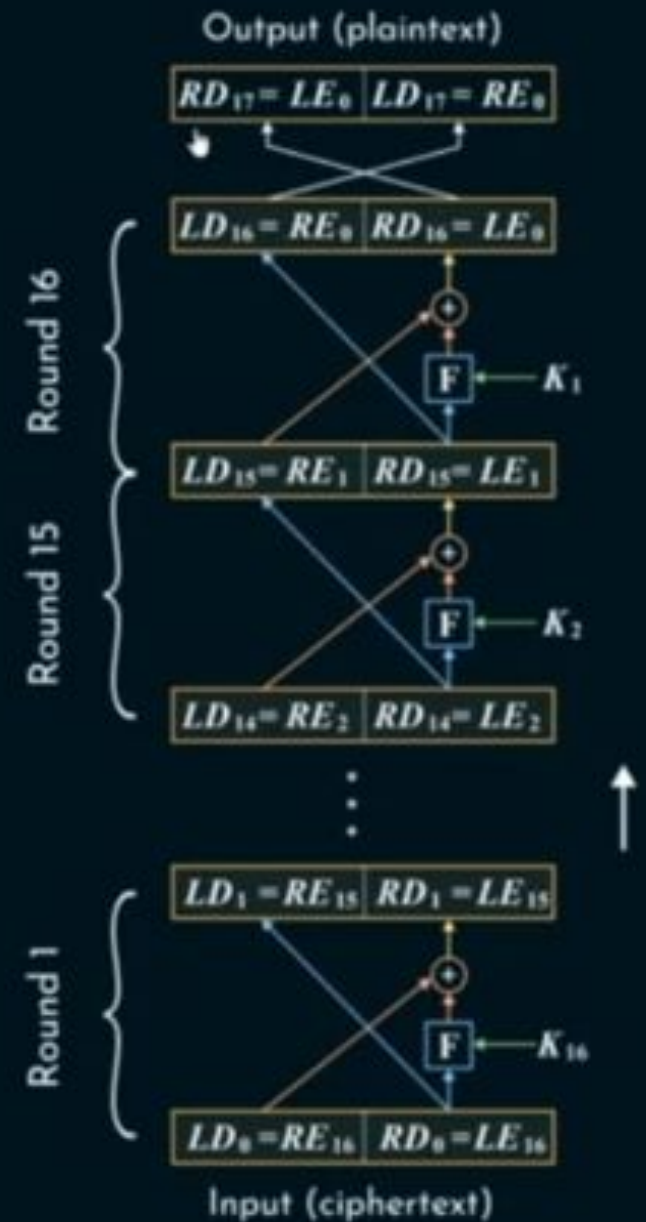
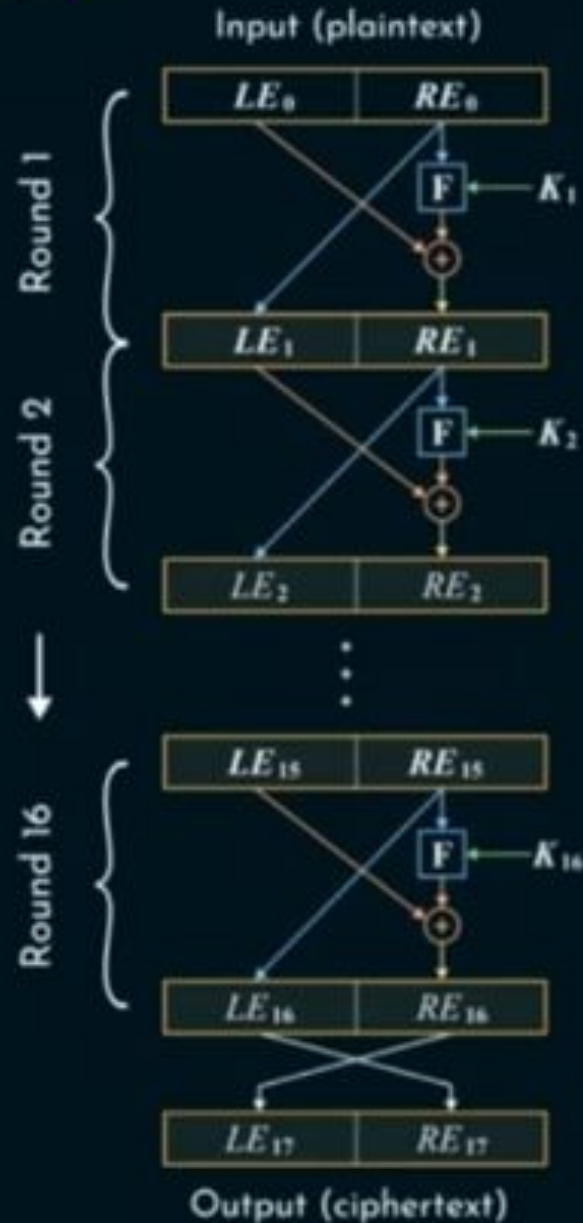
□ **Diffusion:** DES uses bit permutations and swaps across 16 rounds to spread bit changes.

- Feistel networks operate by dividing data into two blocks (left and right) and process these blocks via keyed round functions.

Block cipher



Feistel Structure



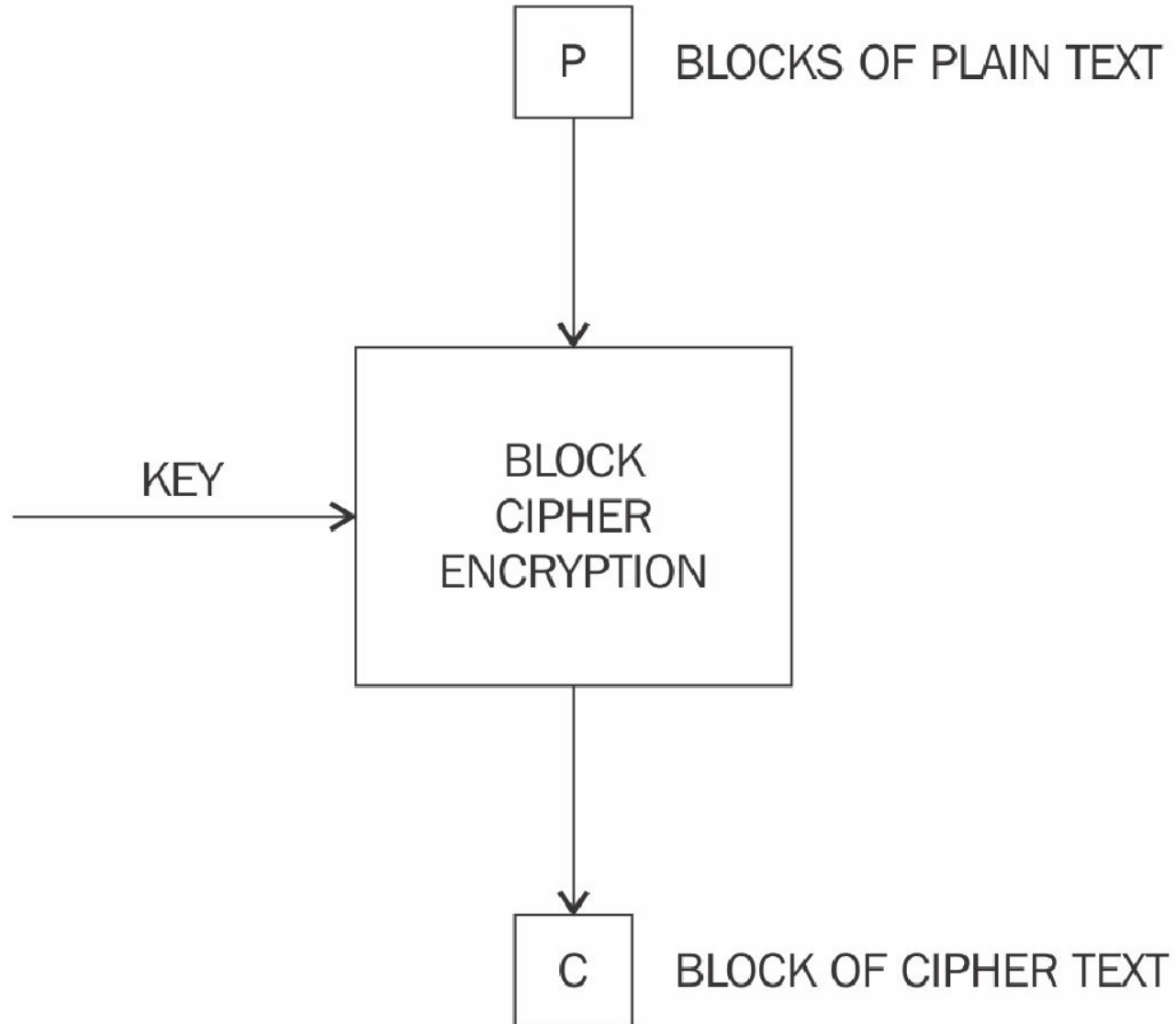
Confusion Operation

- Confusion makes the relationship between the encrypted text and plaintext complex. This is achieved by substitution.
- For example, 'A' in plain text is replaced by 'X' in encrypted text.
- Confusion is required to make finding the encryption key very difficult even if many encrypted and decrypted data pairs are created using the same key.
- A key advantage of using Fiestel cipher is that encryption and decryption operations are almost identical and only require a reversal of the encryption process in order to achieve decryption.

Diffusion Operation

- The diffusion property spreads the plain text statistically over the encrypted data, which ensures that even if a single bit is changed in the input text, it results in changing at least half (on average) of the bits in the cipher text.
- Example: Transposition or permutation.

Operation of block cipher



- Various modes of operation for block ciphers are

- **Electronic Code Book (ECB),**

- **Cipher block chaining (CBC),**

- **Output Feedback Mode (OFB), or Counter mode (CTR).**

- These modes are used to specify the way in which an encryption function would be applied to the plain text.
- First four categories of block cipher encryption modes.
- **Block encryption mode**
- **Keystream generation modes**
- **Message authentication modes**
- **Cryptographic hashes**

Block encryption mode

- In this mode, plaintext is divided into blocks of fixed length depending on the type of cipher used and then the encryption function is applied on each block.

Keystream generation modes

- In this mode, the encryption function generates a keystream that is then XORed with the plaintext stream in order to achieve encryption.

Message authentication modes

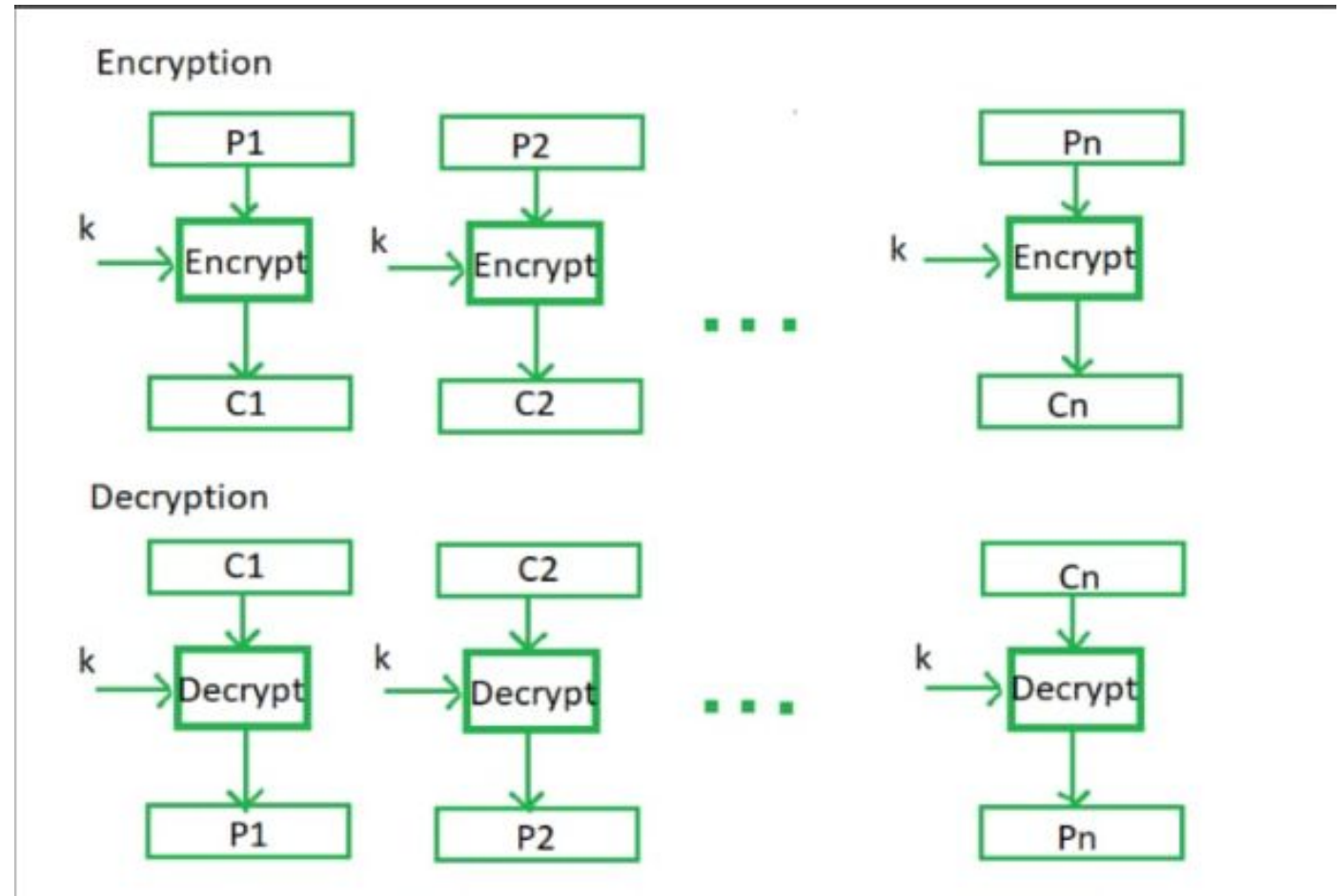
- In this mode, a message authentication code is computed as a result of an encryption function. MAC is basically a cryptographic checksum that provides an integrity service. The most common method to generate MAC using block ciphers is CBC-MAC, where some part of the last block of the chain is used as a MAC.

Cryptographic hashes

- Hash functions are basically used to compress a message to a fixed length digest. In this mode, block ciphers are used as a compression function to produce a hash of plain text.

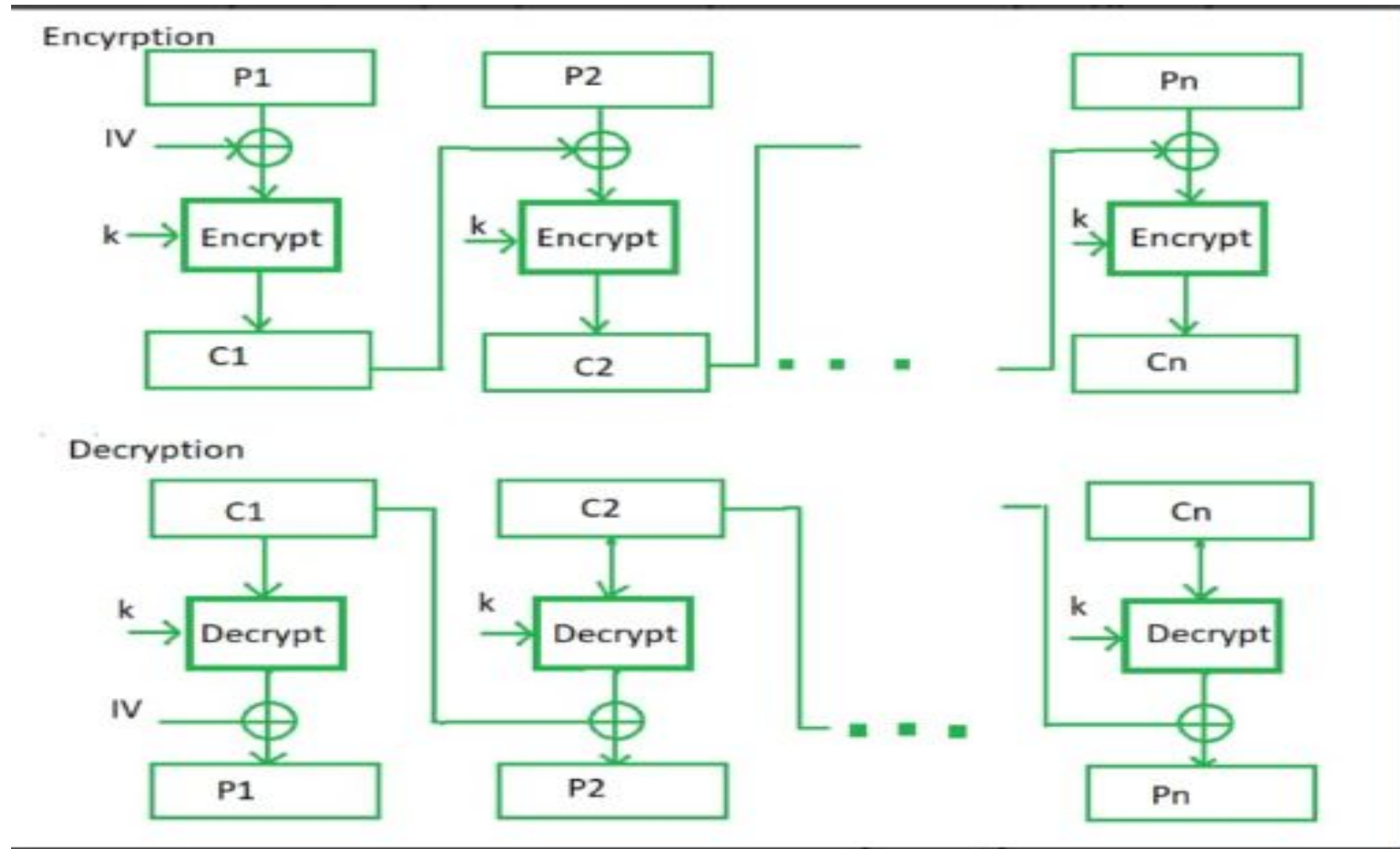
Electronic code book

- This is a basic mode of operation in which the encrypted data is produced as a result of applying the encryption algorithm one by one separately to each block of plain text. This is the simplest mode but should not be used in practice as it is insecure and can reveal information.
- It is used only for short messages



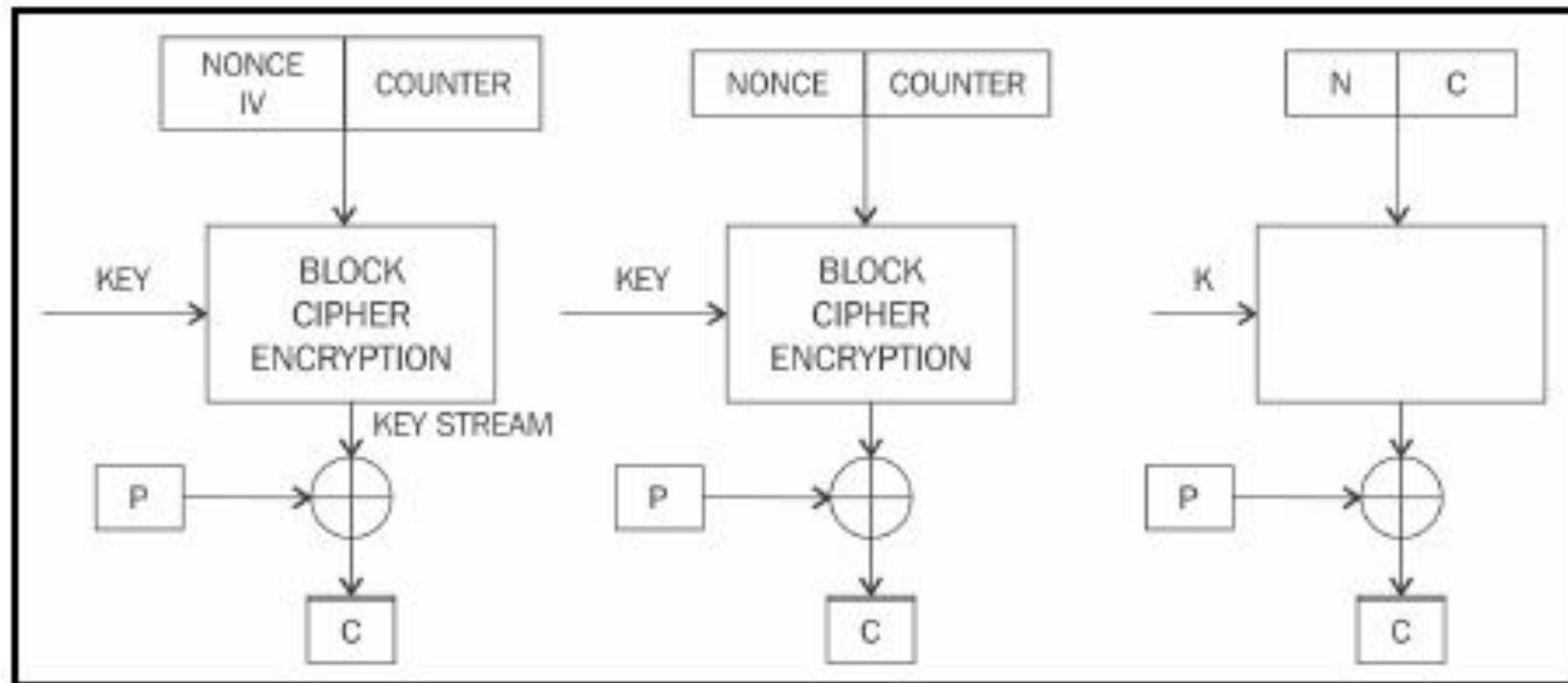
Cipher block chaining

- In this mode, each block of plain text is XORed with the previous encrypted block. The CBC mode uses initialization vector IV to encrypt the first block. It is recommended that IV be randomly chosen:



Counter mode

- The CTR mode effectively uses a block cipher as a stream cipher.
- In this case, a unique nonce is supplied that is concatenated with the counter value in order to produce a **key stream**:



Counter mode

Open SSL Commands

- Electronic code book :

Encryption:

```
openssl enc -aes-128-ecb -e -in plaintext.txt -out encrypted.bin -K  
00112233445566778899AABBCCDDEEFF
```

Decryption:

```
openssl enc -aes-128-ecb -d -in encrypted.bin -out decrypted.txt -K  
00112233445566778899AABBCCDDEEFF
```

- **Cipher Block chaining :**

Encryption:

```
openssl enc -aes-128-cbc -e -in plaintext.txt -out encrypted.bin -K  
00112233445566778899AABBCCDDEEFF -iv  
0102030405060708090A0B0C0D0E0F10
```

Decryption:

```
openssl enc -aes-128-cbc -d -in encrypted.bin -out decrypted.txt -K  
00112233445566778899AABBCCDDEEFF -iv  
0102030405060708090A0B0C0D0E0F10
```

- **Counter Mode :**

Encryption:

```
openssl enc -aes-128-ctr -e -in plaintext.txt -out encrypted.bin -K  
00112233445566778899AABBCCDDEEFF -iv  
0102030405060708090A0B0C0D0E0F10
```

Decryption:

```
openssl enc -aes-128-ctr -d -in encrypted.bin -out decrypted.txt -K  
00112233445566778899AABBCCDDEEFF -iv  
0102030405060708090A0B0C0D0E0F10
```

DES

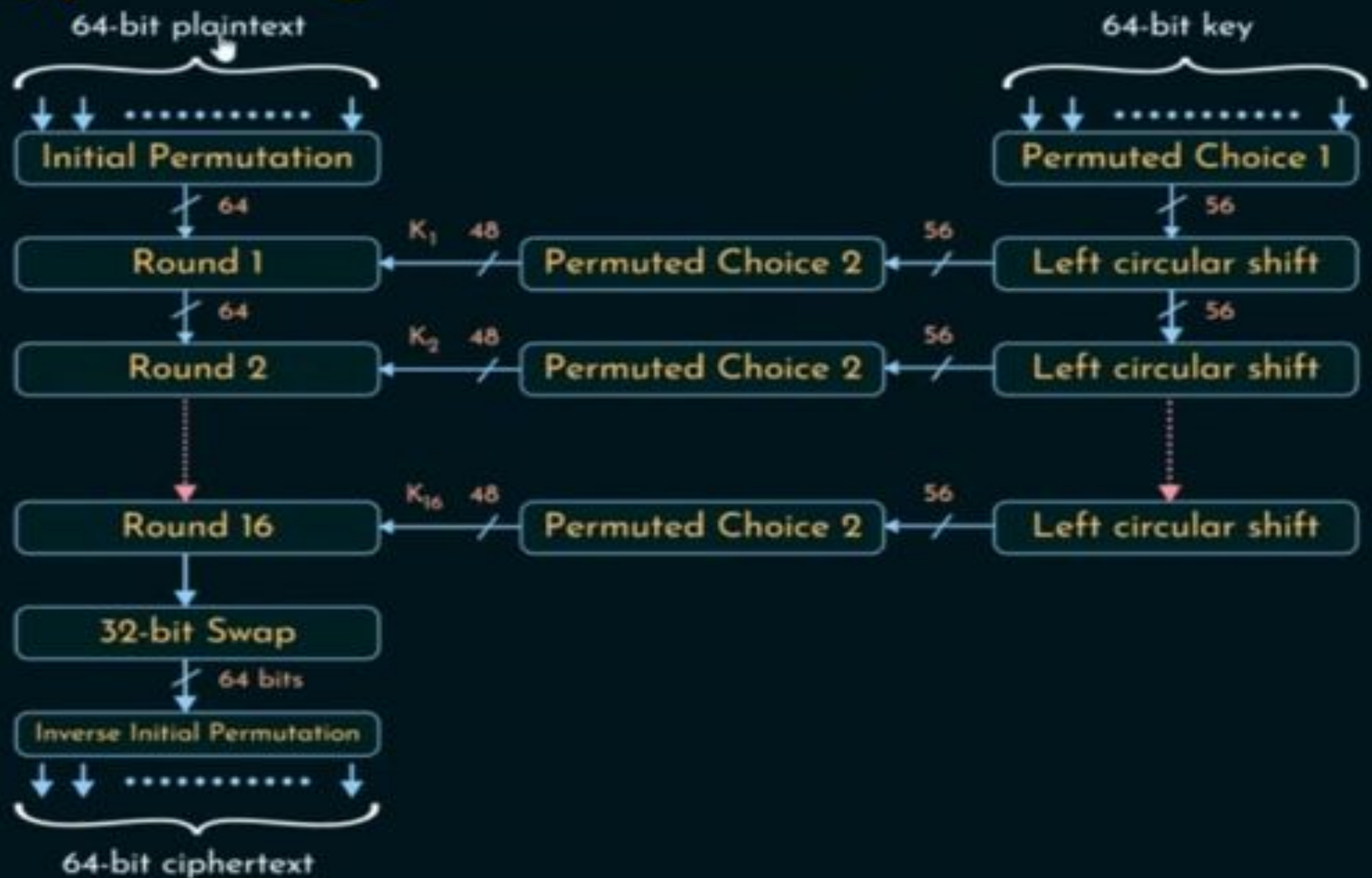
- DES was introduced by the US **National Institute of Standards and Technology (NIST)** as a standard algorithm for encryption and was in main use during 1980s and 1990s.
- In July 1998, **Electronic Frontier Foundation (EFF)** broke DES using a special purpose machine.
- This problem was addressed with the Introduction of **Triple DES (3DES)**, which proposed the usage of a 168-bit key using three 56-bit keys and the same number of executions of the DES algorithm, thus making brute force attacks almost impossible.
- But other limitations, such as slow performance and 64-bit block size.
- DES is a block cipher and encrypts data in blocks of size of 64 bits each, which means 64 bits of plain text go as the Input to DES, which produces 64 bits of ciphertext.
- The key length is 56 bits.

- The initial key consists of 64 bits.
- However, before the DES process even starts, every 8th bit of the key is discarded to produce a 56-bit key. That is bit positions 8, 16, 24, 32, 40, 48, 56, and 64 are discarded.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64

Figure - discarding of every 8th bit of original key

-
- 56 bits are given to Left Circular shift.
 - Left circular shift is one bit shift for rounds, 1,2,9, and 16.
 - Other rounds Left circular shift is two bit shift.
 - 8 bits are dropped and 48 bits are permuted.



Initial Permutation

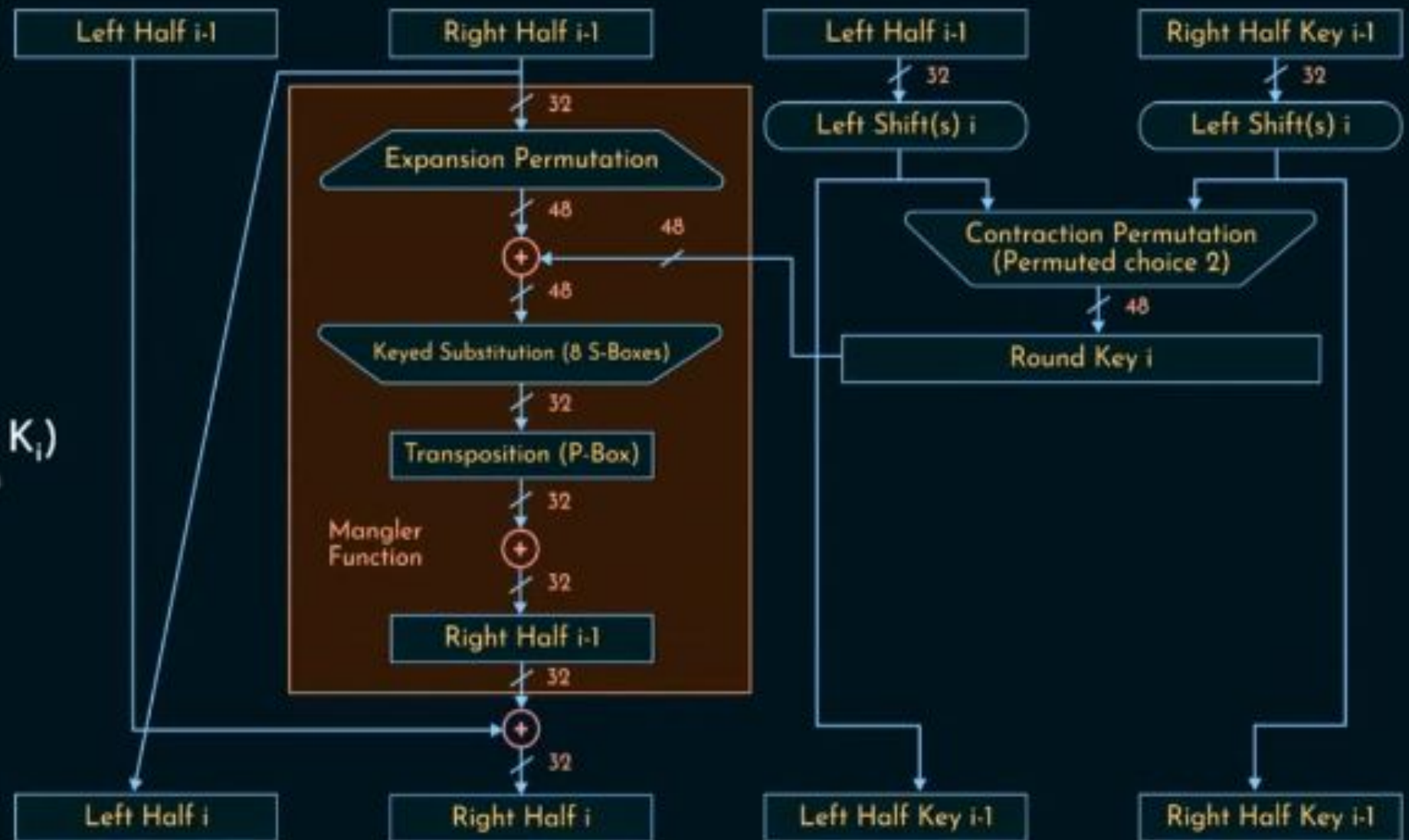
1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64



58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

Single Round of DES Algorithm

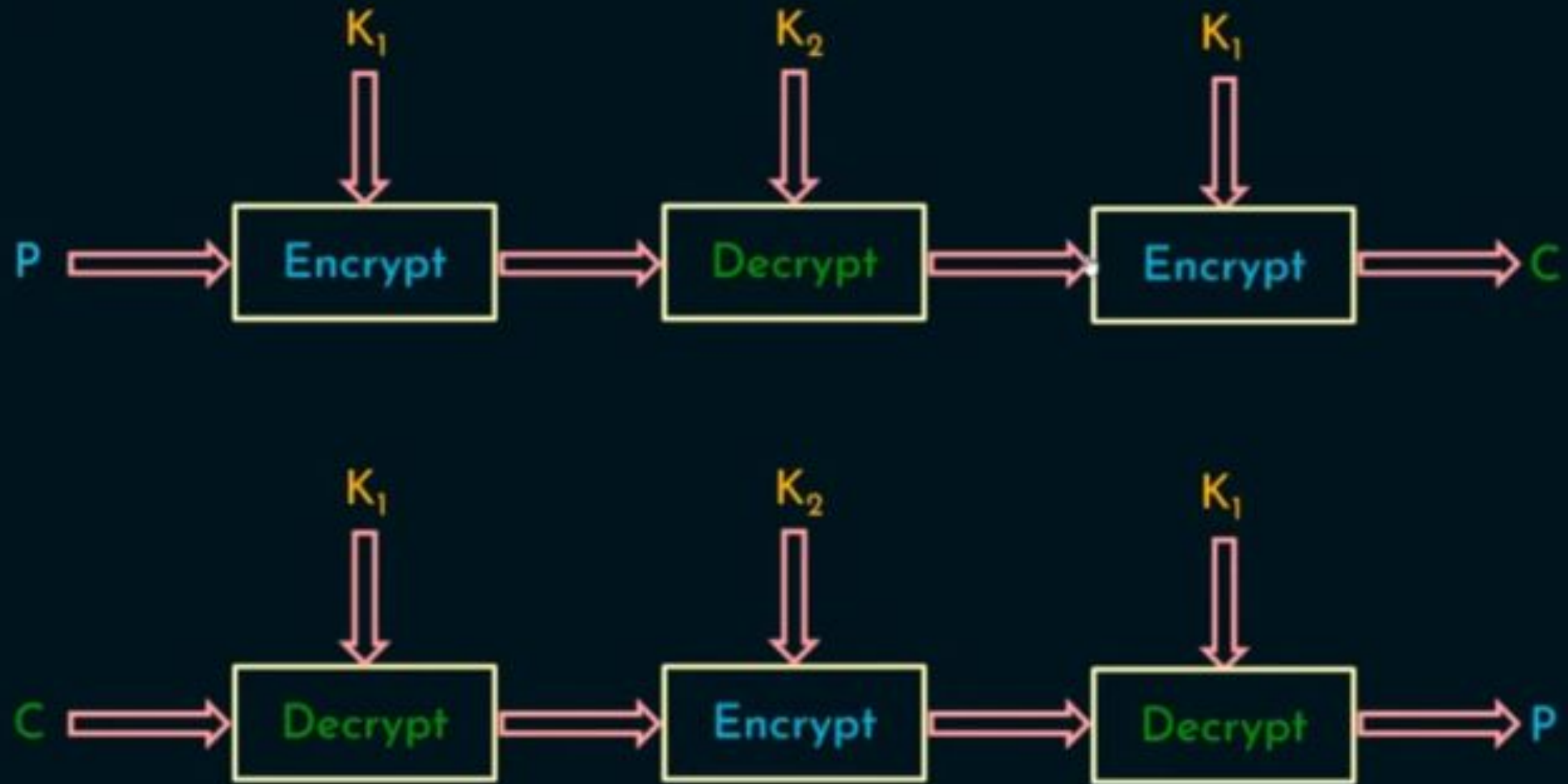
$$L_i = R_{i-1}$$
$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$



DES Decryption



Triple DES

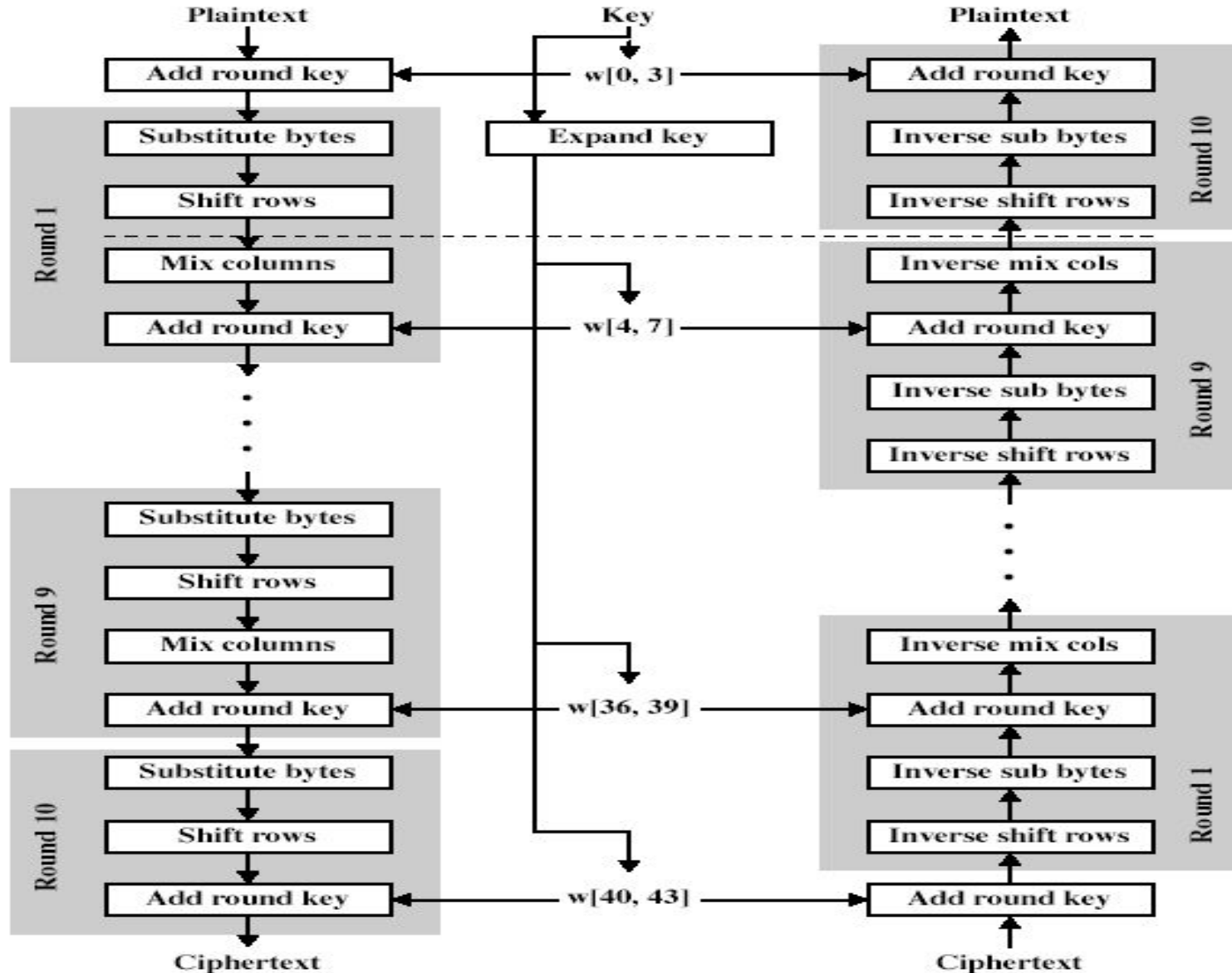


Advanced Encryption Standard (AES)

- 2001, Rijndael
- Block Cipher
- Allows different key of size 128-bit, 192-bit, and 256-bits, but in the AES standard, only a 128-bit block size is allowed.
- However, key sizes of 128-bit, 192-bit, and 256-bit are allowed

Key size	Number of rounds required
128-bit	10 rounds
192-bit	12 rounds
256-bit	14 rounds

Advanced Encryption Standard (AES)



(a) Encryption

(b) Decryption

Advanced Encryption Standard (AES)

- Four operations are performed in four stages in order to encrypt the input.
- These stages are AddRoundKey, SubBytes, ShiftRows, and MixColumns:
 1. In the AddRoundKey step, the state array is XORed with a subkey, which is derived from the master key.
 2. This is the substitution step where a lookup table (S-box) is used to replace all bytes of the state array.
 3. This step is used to shift each row except the first one in the state array to the left in a cyclic and incremental manner.
 4. Finally, all bytes are mixed in this step in a linear fashion column-wise.

Advanced Encryption Standard (AES)

